sized_hdpe_500N
Hybrid Rocket Motor Design Report (HyProp)

Generated by HyProp

July 16, 2026

1 Mission Requirements

Required thrust: 500 N; total impulse: 4e+03 N s; design altitude: 1e+03 m; safety factor: 1.5.

2 Inputs and Assumptions

Oxidizer: N2O at 293 K (self-pressurized). Fuel: HDPE. Injector: 4×1.65 mm, C_d=0.6 (SPI). Efficiencies: η_{c^*}=0.95, nozzle extra η=0.9673, divergence half-angle 15.0°.

3 Grain and Chamber Geometry

Geometry: cylindrical; length 508 mm; outer diameter 43.1 mm; port parameters: 'initial$_{d}iameter'_{m}$: $0.025997243922216768.Tank : 2.4L, 1.66kgoxidizer.$

4 Transient Simulation

Burn time 17 s; total impulse 4.08e+03 N s; peak thrust 604 N; peak chamber pressure 38.2 bar; propellant used 2.05 kg; mean delivered I_{sp} 203 s. Events: liquid runout at 7.24 s; pressure decay at 17.04 s.

5 Representative Operating Points

The plain time average is misleading for blowdown motors: it understates thrust by 51% and chamber pressure by 48% versus the impulse-weighted point. All design modules therefore use the weighted points below. Time share needed to deliver impulse fractions: 50

Table 1: Representative operating points vs plain time average (SI units).

	pc_max	pc_max_time_s	thrust_max	thrust_max_time_s	t_c_max	t_c_max_time_s	cstar	d_port	g_flux	gamma	impulse	isp	m_fuel
peak_performance	3824937.316700	0.000000	604.445500	0.000000	3242.041900	0.000000	1533.595100	0.026000	426.615500	1.249800	12.088900	235.490700	0.458600
impulse_weighted	NaN	NaN	NaN	NaN	NaN	NaN	1487.907000	0.031800	254.072700	1.258100	2049.346000	220.504200	0.324200
heat_load	NaN	NaN	NaN	NaN	NaN	NaN	1533.595100	0.026000	426.615500	1.249800	12.088900	235.490700	0.458600
mass_flow	NaN	NaN	NaN	NaN	NaN	NaN	1454.966800	0.032700	227.873700	1.258600	2283.116500	202.787500	0.301100
median_impulse	NaN	NaN	NaN	NaN	NaN	NaN	1506.145000	0.031800	249.866500	1.258700	2049.618200	227.058800	0.329600
time_average	NaN	NaN	NaN	NaN	NaN	NaN	1319.093400	0.036300	122.973700	1.260600	3174.141100	121.275300	0.203400

	m_ox	m_ox_liquid	mdot_fuel	mdot_ox	mdot_total	of	p_tank	pc	rdot	t_c	t_tank	thrust	web
peak_performance	1.661100	1.604200	0.035300	0.226500	0.261700	6.418600	5060128.300900	3824937.316700	0.000900	3242.041900	293.150000	604.445500	0.000000
impulse_weighted	0.887900	0.717300	0.034700	0.188600	0.223300	5.360700	4122378.712900	3196133.940000	0.000700	2952.310600	282.782400	493.747700	0.002900
heat_load	1.661100	1.604200	0.035300	0.226500	0.261700	6.418600	5060128.300900	3824937.316700	0.000900	3242.041900	293.150000	604.445500	0.000000
mass_flow	0.793100	0.631600	0.032800	0.170900	0.203700	4.915500	3771809.826100	2901048.817000	0.000700	2797.979600	273.495200	444.096100	0.003300
median_impulse	0.891300	0.698200	0.035700	0.198100	0.233800	5.545700	4240629.483300	3355019.870700	0.000700	3033.986500	285.525400	520.518600	0.002900
time_average	0.424300	0.311900	0.023900	0.096500	0.120400	2.999400	2249226.717400	1669919.114600	0.000500	2165.917900	228.286000	239.526300	0.005100

Table 2: Burn phases: liquid / transition / gas.

	t_start_s	t_end_s	impulse_ns	impulse_share	time_share	mean_thrust	mean_pc	mean_of	mean_isp	mean_cstar	mean_gamma	mean_t_c
phase												
liquid	0.000000	7.220000	3753.757500	0.920800	0.424200	519.905800	3350457.261800	5.561600	226.987300	1506.745400	1.258500	3038.444200
transition	7.240000	7.480000	52.018200	0.012800	0.014100	224.065800	1617384.014200	3.325200	179.959300	1344.440700	1.260400	2220.515900
gas	7.500000	17.020000	258.502500	0.063400	0.559300	27.296300	396599.261400	1.046700	39.442700	1176.051300	1.262100	1502.630300

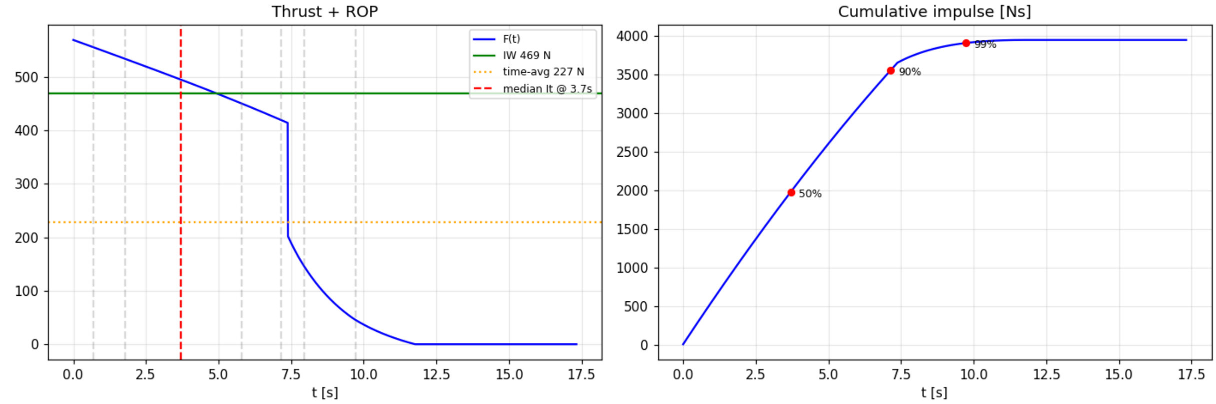
5.1 Recommendations

- Nozzle design and reported performance: **impulse-weighted** point (each instant weighted by its contribution to total impulse).
- Thermal/erosion analysis: **heat-load** point (maximum of the Bartz $\sim P_c^{0.8}$ proxy).
- Structural sizing and materials: **peak-performance** envelope (P_c=38.2 bar, T_c=3242 K).
- Injector and feed system: **mass-flow-weighted** point.

6 Nozzle and Performance

Throat diameter 11.7 mm; expansion ratio 5.02; impulse-weighted I_{sp} 221 s, c^* 1.49e+03 m/s, O/F 5.36.

7 Figures



8 Validation Note

Models validated against NASA-CEA-class thermochemistry (Cantera cross-check), HRAP transient benchmark (prusak_v2) and ESDU 91022/CoolProp N2O properties; see repository README. Empirical coefficients (a, n, C_d, η) require confirmation by fire tests before design freeze.